PADHAKU
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CHAPTER 3

Number Play

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
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⭐ Math Talk · Introduction

? **List five different situations in which numbers are used.**

💡 SAMPLE ANSWER

- Reading the time on a clock
- Dates on a calendar
- Counting money while shopping
- Marks in a test
- Measuring height and weight

⭐ 3.1 Numbers can Tell us Things

? **Children standing in a line each say a number. What do you think these numbers mean?**

✔ ANSWER

Each child says **the number of taller neighbours** they have (0, 1 or 2).

1 **Can the children rearrange themselves so that the children standing at the ends say '2'?**

✔ ANSWER

No. A child at an end has only one neighbour, so they can say at most 1.

2 **Can we arrange the children in a line so that all would say only 0s?**

✔ ANSWER

No (if their heights are different).

Of any two neighbours, the shorter one has a taller neighbour, so they say at least 1.

Only if all children had the same height could everyone say 0.

3 **Can two children standing next to each other say the same number?**

✔ ANSWER

Yes. Example: 3 children in increasing order of height say 1, 1, 0. The first two are neighbours and both say 1.

(But two neighbours can never both say 0 or both say 2.)

4 **There are 5 children in a group, all of different heights. Can they stand such that four of them say '1' and the last one says '0'? Why or why not?**

✔ ANSWER

Yes. Stand them from shortest to tallest.

Each of the first four has a taller child next to them → says 1. The tallest at the end says 0.

Numbers said: 1, 1, 1, 1, 0.

5 **For this group of 5 children, is the sequence 1, 1, 1, 1, 1 possible?**

✔ ANSWER

No. The tallest child has no taller neighbour, so the tallest always says 0.

6 **Is the sequence 0, 1, 2, 1, 0 possible? Why or why not?**

✔ ANSWER

Yes. Put the shortest child in the middle and the tallest ones at the ends.

Heights (1 = shortest): 5, 3, 1, 2, 4 → they say 0, 1, 2, 1, 0.

7 **How would you rearrange the five children so that the maximum number of children say '2'?**

✔ ANSWER

A child saying 2 cannot be at an end, and two neighbours cannot both say 2.

So only positions 2 and 4 can say 2 → at most **2 children**.

Example (1 = shortest): 3, 1, 4, 2, 5 → they say 0, 2, 0, 2, 0.

Figure it Out · 3.2 Supercells

? **Why are some numbers coloured?**

✔ ANSWER

A cell is coloured (a **supercell**) if its number is larger than the numbers in all its adjacent cells.

1 **Colour or mark the supercells in the table below.**

6828	670	9435	3780	3708	7308	8000	5583	52

✔ ANSWER

- 6828 > 670 → supercell

- 9435 > 670 and 3780 → supercell
- 8000 > 7308 and 5583 → supercell

Supercells: **6828, 9435, 8000**

2 **Fill the table with only 4-digit numbers such that the supercells are exactly the coloured cells.**

Given: 5346 (1st cell), 1258 (4th cell), 9635 (8th cell)

💡 **SAMPLE ANSWER**

Make each coloured cell bigger than both its neighbours, and each uncoloured cell smaller than at least one neighbour.

One way (supercells: 2nd, 4th and 9th cells):

5346	5347	1000	1258	1100	1200	1300	9635	9636

Supercells: 5347, 1258, 9636. (check the picture in your book)

3 **Fill a table of 9 cells such that we get as many supercells as possible. Use numbers between 100 and 1000 without repetitions.**

💡 **SAMPLE ANSWER**

Put big and small numbers alternately, starting with a big one:

200	100	300	110	400	120	500	130	600

Supercells: 200, 300, 400, 500, 600

4 **Out of the 9 numbers, how many supercells are there in the table above?**

✔ **ANSWER**

5

5 **Find out how many supercells are possible for different numbers of cells. Do you notice any pattern? What is the method to fill a table to get the maximum number of supercells?**

✔ ANSWER

Cells	1	2	3	4	5	6	7	8	9
Max supercells	1	1	2	2	3	3	4	4	5

- Even number of cells: maximum = cells ÷ 2
- Odd number of cells: maximum = (cells + 1) ÷ 2

Method: start with a big number in the first cell, then put small and big numbers alternately.

6 **Can you fill a supercell table without repeating numbers such that there are no supercells? Why or why not?**

✔ ANSWER

No. The cell with the largest number is bigger than all its neighbours, so it is always a supercell.

7 **Will the cell having the largest number in a table always be a supercell? Can the cell having the smallest number in a table be a supercell? Why or why not?**

✔ ANSWER

- Largest number: **yes, always** a supercell (it is bigger than every neighbour).
- Smallest number: **no, never** (its neighbours are all bigger than it).

8 **Fill a table such that the cell having the second largest number is not a supercell.**

💡 **SAMPLE ANSWER**

Put the second largest number next to the largest number:

100	200	300	400	500	600	700	900	800

800 (second largest) is next to 900, so it is not a supercell.

9 **Fill a table such that the cell having the second largest number is not a supercell but the second smallest number is a supercell. Is it possible?**

💡 **SAMPLE ANSWER**

Yes, it is possible.

200	100	300	400	500	600	700	900	800

- 200 (second smallest) has only one neighbour, 100, so it is a supercell.
- 800 (second largest) is next to 900, so it is not a supercell.

10 **Make other variations of this puzzle and challenge your classmates.**

💡 **SAMPLE ANSWER**

- Fill 9 cells so that there are exactly 3 supercells.
- Fill 10 cells with 4-digit even numbers to get the most supercells.
- Fill a 3 × 3 grid so that only the corner cells are supercells.

⭐ 3.2 Supercells with more rows (Table 2)

? **Complete Table 2 with 5-digit numbers whose digits are 1, 0, 6, 3 and 9 in some order. Only a coloured cell should have a number greater than all its neighbours.**

Given: Row 1: _, 96,301, 36,109, _ · Row 2: _, 13,609, 60,319, 19,306 · Row 3: _, _, 60,193, _ · Row 4: _, 10,963, _, _

💡 **SAMPLE ANSWER**

One way:

Col 1	Col 2	Col 3	Col 4
96,310	96,301	36,109	39,160
96,103	13,609	60,319	19,306
13,906	10,396	60,193	60,931
10,369	10,963	10,936	69,031

Supercells here: 96,310, 39,160, 60,319, 10,963, 69,031. Make sure they match the coloured cells (check the picture in your book).

a **The biggest number in the table is ____.**

✔ **ANSWER**

96,310 (the largest number that can be made with 1, 0, 6, 3, 9)

b **The smallest even number in the table is ____.**

💡 **SAMPLE ANSWER**

10,396 (in the table above)

c **The smallest number greater than 50,000 in the table is ____.**

💡 SAMPLE ANSWER

60,193 (in the table above)

d **Put commas appropriately after the thousands digit.**

✔ ANSWER

Write each number like 96,310 – a comma after the thousands digit (two digits from the left in a 5-digit number).

🧮 Figure it Out · 3.3 Patterns of Numbers on the Number Line

? **Place these numbers on a number line from 1000 to 10,000: 2180, 2754, 1500, 3600, 9950, 9590, 1050, 3050, 5030, 5300 and 8400.**

✔ ANSWER

In increasing order: 1050, 1500, 2180, 2754, 3050, 3600, 5030, 5300, 8400, 9590, 9950

- 1050 just after 1000; 1500 halfway between 1000 and 2000
- 2180 a little after 2000; 2754 about three-quarters of the way to 3000
- 3050 just after 3000; 3600 a little past halfway to 4000
- 5030 just after 5000; 5300 a bit further
- 8400 a little before halfway from 8000 to 9000
- 9590 just past halfway to 10,000; 9950 very close to 10,000

a **Identify the numbers marked and label the remaining positions: number line with 2010 and 2020 marked.**

💡 **SAMPLE ANSWER**

Marks go up by 5:

1990, 1995, 2000, 2005, **2010**, 2015, **2020**, 2025, 2030, 2035

(check the picture in your book)

b **Number line with 9996 and 9997 marked.**

💡 **SAMPLE ANSWER**

Marks go up by 1:

9993, 9994, 9995, **9996**, **9997**, 9998, 9999, 10,000, 10,001, 10,002

(check the picture in your book)

c **Number line with 15,077, 15,078 and 15,083 marked.**

💡 **SAMPLE ANSWER**

Marks go up by 1:

15,077, **15,078**, 15,079, 15,080, 15,081, 15,082, **15,083**, 15,084, 15,085, 15,086

(check the picture in your book)

d **Number line with 86,705 and 87,705 marked.**

💡 **SAMPLE ANSWER**

Marks go up by 1000:

83,705, 84,705, 85,705, **86,705**, **87,705**, 88,705, 89,705, 90,705, 91,705, 92,705

(check the picture in your book)

e **Put a circle around the smallest number and a box around the largest number in each of the sequences above.**

💡 **SAMPLE ANSWER**

The number at the left end is the smallest; the one at the right end is the largest.

- a: circle 1990, box 2035
- b: circle 9993, box 10,002
- c: circle 15,077, box 15,086
- d: circle 83,705, box 92,705

⭐ 3.4 Playing with Digits

? **Find out how many numbers have two digits, three digits, four digits, and five digits.**

✔ **ANSWER**

1-digit	2-digit	3-digit	4-digit	5-digit
9	90	900	9,000	90,000

Example: 2-digit numbers are 10 to 99 → 99 − 9 = **90**.

🧮 Figure it Out · 3.4 Digit sums of numbers

1 (a) **Digit sum 14: Write other numbers whose digits add up to 14.**

💡 **SAMPLE ANSWER**

59, 68, 77, 86, 95, 149, 266, 545, 707, 1229, 5009

1 (b) **What is the smallest number whose digit sum is 14?**

✔ ANSWER

A 1-digit number can have digit sum at most 9, so we need 2 digits.

Keep the tens digit smallest: 5 + 9 = 14 → **59**

1 (c) **What is the largest 5-digit number whose digit sum is 14?**

✔ ANSWER

Put the biggest digits first: 9 + 5 = 14, rest zeros → **95,000**

1 (d) **How big a number can you form having the digit sum of 14? Can you make an even bigger number?**

✔ ANSWER

There is no biggest number. Adding zeros keeps the digit sum 14:

95,000 → 9,50,000 → 95,00,000 → ...

So we can always make an even bigger number.

2 **Find out the digit sums of all the numbers from 40 to 70. Share your observations.**

✔ ANSWER

- 40 to 49: 4, 5, 6, 7, 8, 9, 10, 11, 12, 13
- 50 to 59: 5, 6, 7, 8, 9, 10, 11, 12, 13, 14
- 60 to 69: 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
- 70: 7

Observations:

- Within each ten, the digit sum goes up by 1 each time.
- From 49 to 50 (and 59 to 60, 69 to 70) it drops by 8.
- Many numbers share a digit sum, e.g. 49, 58, 67 all have 13.
- Smallest is 4 (40), largest is 15 (69).

3 **Calculate the digit sums of 3-digit numbers whose digits are consecutive (for example, 345). Do you see a pattern? Will this pattern continue?**

✔ ANSWER

123 → 6, 234 → 9, 345 → 12, 456 → 15, 567 → 18, 678 → 21, 789 → 24

Pattern: each sum is 3 more than the previous one; each sum = 3 × middle digit.

The pattern **stops at 789**: there is no digit after 9, so 89(10) is not possible.

Digit Detectives **Among the numbers 1–100, how many times will the digit '7' occur? Among the numbers 1–1000?**

✔ ANSWER

1–100:

- 7 in units place: 7, 17, ..., 97 → 10 times
- 7 in tens place: 70–79 → 10 times

Total = **20 times**

1–1000:

- units place: 100, tens place: 100, hundreds place (700–799): 100

Total = **300 times**

⭐ 3.5 Pretty Palindromic Patterns

? **Write all possible 3-digit palindromes using the digits 1, 2, 3.**

✔ ANSWER

First and last digits are the same; middle can be any of 1, 2, 3:

111, 121, 131, 212, 222, 232, 313, 323, 333 → **9 palindromes**

? **Reverse-and-add: try the procedure for some 2-digit numbers. Are there numbers for which you do not reach a palindrome at all?**

💡 SAMPLE ANSWER

- 12 + 21 = **33**
- 47 + 74 = **121**
- 39 + 93 = 132 → 132 + 231 = **363**
- 89 takes many rounds (24 steps).

For 2-digit numbers we always reach a palindrome. For 3-digit numbers nobody knows; 196 is suspected to never give one.

Explore **Will reversing and adding numbers repeatedly, starting with a 2-digit number, always give a palindrome?**

✔ ANSWER

Yes, every 2-digit number reaches a palindrome (sometimes after many steps).

Puzzle **I am a 5-digit palindrome. I am an odd number. My 't' digit is double of my 'u' digit.**

My ‘h’ digit is double of my ‘t’ digit. Who am I?

✔ ANSWER

h = 2 × t = 4 × u, and h ≤ 9, so u is 1 or 2.

The number is odd, so u = 1 → t = 2, h = 4.

Palindrome: tth = u = 1, th = t = 2.

Number: **12,421** – Twelve thousand four hundred twenty-one

⭐ 3.6 The Magic Number of Kaprekar

? **Complete the last step:**

A = 8632, B = 2368, C = 6264 → A = 6642, B = 2466, C = 4176 → A = 7641, B = 1467, C = 6174 → A = ?, B = ?, C = ?

What happens if we continue doing this?

✔ ANSWER

Digits of 6174 → A = **7641**, B = **1467**

C = 7641 – 1467 = **6174**

We get 6174 again and again – it keeps repeating.

Explore **Take different 4-digit numbers and try carrying out these steps. Find out what happens.**

💡 SAMPLE ANSWER

Example: 4321

4321 – 1234 = 3087

8730 – 0378 = 8352

8532 – 2358 = 6174

We always reach **6174** (Kaprekar constant).

? **Carry out these same steps with a few 3-digit numbers. What number will start repeating?**

✔ ANSWER

Example: 321

321 − 123 = 198

981 − 189 = 792

972 − 279 = 693

963 − 369 = 594

954 − 459 = 495

954 − 459 = 495 ...

495 starts repeating.

⭐ 3.7 Clock and Calendar Numbers

? **On a 12-hour clock, find all possible times of each type: 4:44, 10:10, 12:21.**

✔ ANSWER

- Like 4:44 (all digits same): 1:11, 2:22, 3:33, 4:44, 5:55, 11:11
- Like 10:10 (minutes same as hours): 1:01, 2:02, 3:03, ..., 9:09, 10:10, 11:11, 12:12
- Like 12:21 (palindromes): 1:01, 1:11, 1:21, 1:31, 1:41, 1:51, and the same for hours 2 to 9 (6 each = 54 times), plus 10:01, 11:11, 12:21 → **57** palindromic times

? **Manish's birthday is 20/12/2012 (digits 2, 0, 1, 2 repeat). Find some other dates of this form from the past.**

✔ ANSWER

20/01/2001, 20/02/2002, 20/03/2003, ..., 20/09/2009, 20/10/2010, 20/11/2011

Also: 10/10/1010, 11/11/1111, 12/12/1212, 19/11/1911

? **Meghana's birthday 11/02/2011 reads the same from both sides. Find all possible dates of this form from the past.**

✔ ANSWER

In this century (up to 2026):

10/02/2001, 20/02/2002, 01/02/2010, 11/02/2011, 21/02/2012, 02/02/2020, 12/02/2021, 22/02/2022

(Older ones exist too, e.g. 01/01/1010, 10/11/1101.)

Try This **Will any year's calendar repeat again after some years? Will all dates and days in a year match exactly with that of another year?**

✔ ANSWER

Yes. Each normal year moves the days by 1, a leap year by 2. When the total shift is 7 days, the calendar repeats.

- A normal year repeats after 6 or 11 years.
- A leap year repeats after 28 years.

Example: 2015, 2026 and 2037 have exactly the same calendar.

Figure it Out · 3.7 Clock and Calendar Numbers

1 (a) **Pratibha's digits 4, 7, 3, 2 give 7432 − 2347 = 5085 and 7432 + 2347 = 9779. Choose 4 digits to make the difference between the largest and smallest numbers greater than 5085.**

💡 SAMPLE ANSWER

Digits 9, 5, 2, 1: 9521 − 1259 = **8262** > 5085

1 (b) **... the difference less than 5085.**

💡 SAMPLE ANSWER

Digits 5, 4, 3, 2: 5432 − 2345 = **3087** < 5085

1 (c) **... the sum of the largest and smallest numbers greater than 9779.**

💡 SAMPLE ANSWER

Digits 9, 8, 7, 6: 9876 + 6789 = **16,665** > 9779

1 (d) **... the sum of the largest and smallest numbers less than 9779.**

💡 SAMPLE ANSWER

Digits 4, 3, 2, 1: 4321 + 1234 = **5555** < 9779

2 **What is the sum of the smallest and largest 5-digit palindrome? What is their difference?**

✔ ANSWER

Smallest 5-digit palindrome = 10,001

Largest 5-digit palindrome = 99,999

Sum = 10,001 + 99,999 = **1,10,000**

Difference = 99,999 − 10,001 = **89,998**

3 **The time now is 10:01. How many minutes until the clock shows the next palindromic time? What about the one after that?**

✔ ANSWER

Next palindromic time: 11:11

10:01 → 11:11 = 1 hour 10 minutes = **70 minutes**

The one after that: 12:21

11:11 → 12:21 = 70 minutes, so **140 minutes** from 10:01

4 **How many rounds does the number 5683 take to reach the Kaprekar constant?**

✔ ANSWER

1. 8653 − 3568 = 5085
2. 8550 − 0558 = 7992
3. 9972 − 2799 = 7173
4. 7731 − 1377 = 6354
5. 6543 − 3456 = 3087
6. 8730 − 0378 = 8352
7. 8532 − 2358 = 6174

7 rounds

(If you do not allow 0 at the start of the smallest number, e.g. 5058 instead of 0558, it takes 8 rounds.)

⭐ 3.8 Mental Math

? **Middle numbers: 25,000, 400, 13,000, 1,500, 60,000 (each can be used many times).**

Make the side numbers: 38,800, 28,000, 61,600, 31,000, 3,400, 63,000, 19,500, 20,900.

✔ ANSWER

- 38,800 = 25,000 + 400 × 2 + 13,000 (given)
- 3,400 = 1,500 + 1,500 + 400 (given)
- 28,000 = 25,000 + 1,500 + 1,500
- 61,600 = 60,000 + 400 × 4
- 31,000 = 25,000 + 1,500 × 4
- 63,000 = 60,000 + 1,500 × 2
- 19,500 = 13,000 + 1,500 × 3 + 400 × 5
- 20,900 = 13,000 + 1,500 × 5 + 400

Math Talk **Can we make 1,000 using the numbers in the middle? Why not? What about 14,000, 15,000 and 16,000? What thousands cannot be made?**

✔ ANSWER

1,000: **No.** Only 400 is smaller than 1,000, and 400, 800, 1,200 ... skip 1,000.

- 14,000 = 1,500 × 8 + 400 × 5 = 12,000 + 2,000
- 15,000 = 13,000 + 400 × 5
- 16,000 = 13,000 + 1,500 × 2

Only **1,000** cannot be made (2,000 = 400 × 5, 3,000 = 1,500 × 2, and all others can be built from these).

Adding and Subtracting **Using 40,000, 7,000, 300, 1,500, 12,000, 800 (add or subtract), make: 45,000; 5,900; 17,500; 21,400**

💡 SAMPLE ANSWER

- 45,000 = 40,000 + 7,000 − 1,500 − 800 + 300
- 5,900 = 7,000 − 800 − 300

- 17,500 = 12,000 + 7,000 − 1,500
- 21,400 = 12,000 + 7,000 + 1,500 + 300 + 300 + 300

Figure it Out · 3.8 Digits and Operations

1 **Write an example for each scenario whenever possible.**

💡 **SAMPLE ANSWER**

- 5-digit + 5-digit, 5-digit sum more than 90,250: 50,000 + 40,300 = 90,300
- 5-digit + 3-digit = 6-digit sum: 99,999 + 999 = 1,00,998
- 4-digit + 4-digit = 6-digit sum: **not possible** (largest is 9,999 + 9,999 = 19,998)
- 5-digit + 5-digit = 6-digit sum: 60,000 + 40,000 = 1,00,000
- 5-digit + 5-digit = 18,500: **not possible** (smallest is 10,000 + 10,000 = 20,000)
- 5-digit − 5-digit, difference less than 56,503: 80,000 − 50,000 = 30,000
- 5-digit − 3-digit = 4-digit difference: 10,000 − 999 = 9,001
- 5-digit − 4-digit = 4-digit difference: 12,000 − 2,500 = 9,500
- 5-digit − 5-digit = 3-digit difference: 50,999 − 50,000 = 999
- 5-digit − 5-digit = 91,500: **not possible** (largest is 99,999 − 10,000 = 89,999)

2 (a) **Always, sometimes or never? 5-digit number + 5-digit number gives a 5-digit number**

✔ ANSWER

Only sometimes true. 10,000 + 20,000 = 30,000 (5-digit), but 60,000 + 50,000 = 1,10,000 (6-digit).

2 (b) **4-digit number + 2-digit number gives a 4-digit number**

✔ ANSWER

Only sometimes true. 1,000 + 10 = 1,010, but 9,999 + 99 = 10,098 (5-digit).

2 (c) **4-digit number + 2-digit number gives a 6-digit number**

✔ ANSWER

Never true. The biggest sum is 9,999 + 99 = 10,098, only 5 digits.

2 (d) **5-digit number − 5-digit number gives a 5-digit number**

✔ ANSWER

Only sometimes true. 50,000 − 20,000 = 30,000 (5-digit), but 20,000 − 15,000 = 5,000 (4-digit).

2 (e) **5-digit number − 2-digit number gives a 3-digit number**

✔ ANSWER

Never true. The smallest result is 10,000 − 99 = 9,901, which already has 4 digits.

⭐ 3.9 Playing with Number Patterns

a **Find the sum: 3 rows of four 40s and 2 rows of five 50s.**

✔ ANSWER

40s: 3 × 4 = 12 → 12 × 40 = 480

50s: 2 × 5 = 10 → 10 × 50 = 500

Sum = 480 + 500 = **980**

b **Find the sum of the numbers in figure b.**

💡 SAMPLE ANSWER

Quick way: count how many times each number appears, multiply, then add the results. (check the picture in your book)

c **Find the sum: four rows of eight 32s and four rows of 64s.**

💡 SAMPLE ANSWER

32s: 4 × 8 = 32 → 32 × 32 = 1,024

64s: 4 × 4 = 16 → 16 × 64 = 1,024

Sum = 1,024 + 1,024 = **2,048** (check the picture in your book)

d **Find the sum of the numbers in figure d.**

💡 SAMPLE ANSWER

Count the 64s (or other numbers), multiply by the count and add. (check the picture in your book)

e **Find the sum of the 15s, 25s and 35s in figure e.**

💡 **SAMPLE ANSWER**

Each of 15, 25 and 35 appears 22 times.

Group one 15, one 25 and one 35: 15 + 25 + 35 = 75

Sum = 22 × 75 = **1,650** (check the picture in your book)

f **Find the sum of the numbers in figure f (1000 in the middle, then 500s, 250s, 125s).**

💡 **SAMPLE ANSWER**

- 1000 × 1 = 1,000
- 500 × 4 = 2,000
- 250 × 8 = 2,000
- 125 × 18 = 2,250

Sum = **7,250** (count the 125s in your book; with 16 of them the sum is 7,000)

⭐ 3.10 An Unsolved Mystery — the Collatz Conjecture!

? **Look at the sequences 12, 6, 3, 10, 5, 16, 8, 4, 2, 1 etc. Do you see how these sequences were formed?**

✔ **ANSWER**

If the number is even, take half of it; if it is odd, multiply by 3 and add 1. Repeat.

? **Make some more Collatz sequences starting with your favourite whole numbers. Do you always reach 1?**

💡 SAMPLE ANSWER

- 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1
- 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1
- 9, 28, 14, 7, … , 4, 2, 1

Yes, every number we tried reached 1.

? **Do you believe the conjecture of Collatz that all such sequences will eventually reach 1? Why or why not?**

💡 SAMPLE ANSWER

It seems true – every number tried so far reaches 1.

But no one has proved it for all numbers, so we cannot be sure. It is still unsolved.

Figure it Out · 3.11 Simple Estimation

? **How many students are in your school? About 150? 400? A thousand?**

💡 SAMPLE ANSWER

Example: 3 sections × about 35 students × 8 classes (6–12 and one more) ≈ 800 students.

1 **Steps you would take to walk: (a) to the classroom door (b) across the school ground (c) from**

classroom door to school gate (d) from school to home

💡 SAMPLE ANSWER

- a: about 10–20 steps
- b: about 100–150 steps
- c: about 100–300 steps
- d: for 1 km, about 1,500–2,000 steps

2 **Number of times you blink your eyes or number of breaths you take: (a) in a minute (b) in an hour (c) in a day**

💡 SAMPLE ANSWER

Breaths (about 15 per minute):

- a: about 15
- b: 15 × 60 ≈ 900
- c: 900 × 24 ≈ 21,600

Blinks: about 15–20 a minute, about 1,000 an hour.

3 **Name some objects around you that are: (a) a few thousand in number (b) more than ten thousand in number**

💡 SAMPLE ANSWER

- a: bricks in a classroom wall, pages of all books in the class, leaves on a small tree
- b: grains of rice in a bag, hairs on your head, grains of sand in a heap, leaves on a big tree

⭐ Estimate the answer · 3.11

1 **Number of words in your maths textbook: (a) more than 5000 (b) less than 5000**

✔ ANSWER

(a) More than 5000. About 300 pages × 100 words each ≈ 30,000 words.

2 **Number of students in your school who travel to school by bus: (a) more than 200 (b) less than 200**

💡 SAMPLE ANSWER

Depends on your school. Example: in a school of 800 where about 1 in 5 come by bus, ≈ 160 → **less than 200**.

3 **Roshan wants to buy milk and 3 types of fruit to make fruit custard for 5 people. He estimates the cost to be ₹100. Do you agree?**

💡 SAMPLE ANSWER

No, ₹100 is too low.

1 litre of milk ≈ ₹60–70, and 3 kinds of fruit for 5 people ≈ ₹100 or more.

A better estimate is about ₹200–250.

4 **Estimate the distance between Gandhinagar (Gujarat) and Kohima (Nagaland).**

💡 SAMPLE ANSWER

On the map of India (straight line) about **2,000–2,500 km**; by road it is more, around 3,000 km.

5 **Sheetal is in Grade 6 and says she has spent around 13,000 hours in school till date. Do you agree?**

💡 SAMPLE ANSWER

No.

About 5 hours a day × about 220 school days ≈ 1,100 hours a year.

Classes 1 to 5 plus part of 6 ≈ 6 years → 6 × 1,100 ≈ 6,600 hours.

Even with 2 years of pre-school it is under 9,000 hours, so 13,000 is too high.

6 **Walking at your normal pace (about 5 km per hour), how long would it take to go from: (a) your location to a favourite place nearby (b) to a neighbouring state's capital (c) the southernmost point to the northernmost point of India**

💡 SAMPLE ANSWER

- a: a park 2 km away → about 25 minutes
- b: e.g. Delhi to Jaipur ≈ 280 km → 280 ÷ 5 ≈ 56 hours, about 7 days walking 8 hours a day
- c: about 3,200 km → 3,200 ÷ 5 = 640 hours, about 80 days walking 8 hours a day

7 **Make some estimation questions and challenge your classmates!**

💡 SAMPLE ANSWER

- How many litres of water does your family use in a day?
- How many rotis does your family eat in a month?

- How many hours do you sleep in a year?

⭐ 3.12 Games and Winning Strategies

Game #1 **Game of 21 (add 1, 2 or 3; first to reach 21 wins). Which player can always win? What is the pattern of numbers the winning player should say?**

✔ ANSWER

The **first player** can always win.

Say **1, 5, 9, 13, 17, 21** (go up by 4 each time).

If the other player adds 1, 2 or 3, you add 3, 2 or 1 to make the total 4.

Game #2 **Game of 99 (add a number from 1 to 10; first to reach 99 wins). Which player can always win? What pattern should they say?**

✔ ANSWER

The **second player** can always win.

Say the multiples of 11: **11, 22, 33, 44, 55, 66, 77, 88, 99**.

Whatever the other player adds (1–10), you add enough to make 11.

? **Make your own variations of this game and figure out the winning strategy.**

💡 SAMPLE ANSWER

Example: add 1 to 4; first to 30 wins.

Key numbers go down by 5 from 30: 25, 20, 15, 10, 5, 0.

The first player must start with 1–4, so the second player says 5, 10, 15, 20, 25, 30 and wins.

Figure it Out · 3.12 (end of chapter)

1 **There is only one supercell in this grid. Exchange two digits of one number so that there are 4 supercells.**

16,200	39,344	29,765
23,609	62,871	45,306
19,381	50,319	38,408

✔ ANSWER

Swap the digits 6 and 1 in 62,871 → **12,876**

16,200	39,344	29,765
23,609	12,876	45,306
19,381	50,319	38,408

Supercells: 39,344, 23,609, 45,306, 50,319 → 4 supercells

2 **How many rounds does your year of birth take to reach the Kaprekar constant?**

💡 SAMPLE ANSWER

Example: 2014

1. 4210 − 0124 = 4086
2. 8640 − 0468 = 8172
3. 8721 − 1278 = 7443
4. 7443 − 3447 = 3996
5. 9963 − 3699 = 6264
6. 6642 − 2466 = 4176
7. 7641 − 1467 = 6174

7 rounds. Now try your own year of birth.

3 **We are the group of 5-digit numbers between 35,000 and 75,000 such that all of our digits are odd. Who is the largest? Who is the smallest? Who is closest to 50,000?**

✔ ANSWER

Largest: first digit 7, next odd digit below 5 is 3 → **73,999**

Smallest: **35,111**

Closest to 50,000: below it 39,999 (10,001 away), above it 51,111 (1,111 away) → **51,111**

4 **Estimate the number of holidays you get in a year including weekends, festivals and vacation. Then find the exact number.**

💡 SAMPLE ANSWER

- Sundays: 52
- Second Saturdays etc.: about 12
- Festivals: about 15
- Summer and winter vacations: about 50

Estimate ≈ **130 days**. Check with your school calendar.

5 **Estimate the number of litres a mug, a bucket and an overhead tank can hold.**

💡 SAMPLE ANSWER

- Mug: about 1 litre
- Bucket: about 15–20 litres
- Overhead tank: about 500–1,000 litres

6 **Write one 5-digit number and two 3-digit numbers such that their sum is 18,670.**

✔ ANSWER

18,000 + 300 + 370 = **18,670**

(Another: 17,670 + 500 + 500 = 18,670)

7 **Choose a number between 210 and 390. Create a number pattern similar to those in Section 3.9 that will sum up to this number.**

💡 SAMPLE ANSWER

Number chosen: 300

25	50	25
50	0	50
25	50	25

Sum = 4 × 25 + 4 × 50 = 100 + 200 = **300**

8 **Recall the sequence of Powers of 2 from Chapter 1. Why is the Collatz conjecture correct for all the starting numbers in this sequence?**

✔ ANSWER

Powers of 2: 1, 2, 4, 8, 16, 32, 64, ...

Each one is even (except 1), so we keep halving: 64 → 32 → 16 → 8 → 4 → 2 → 1.

We never get an odd number before 1, so we **always reach 1**.

9 **Check if the Collatz Conjecture holds for the starting number 100.**

✔ ANSWER

100, 50, 25, 76, 38, 19, 58, 29, 88, 44, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1

Yes, it reaches 1.

10 **Starting with 0, players alternate adding numbers between 1 and 3. The first person to reach 22 wins. What is the winning strategy now?**

✔ ANSWER

Key numbers go down by 4 from 22: 22, 18, 14, 10, 6, 2.

The **first player** wins: start by saying **2**, then always say 6, 10, 14, 18, 22.

If the other player adds 1, 2 or 3, add 3, 2 or 1.

🎉 **Chapter 3 done! Revise the answers once more tomorrow.**